THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** second	1 minute = 60 seconds.
2	**(c)** distance travelled per unit time	Speed = distance/time.
3	**(d)** odometer	Shows kilometres covered.
4	**(d)** 3600 seconds	60 x 60.
5	**(c)** metre per second	m/s.
6	**(d)** speedometer	Shows km/h.
7	**(b)** time period	Time for one to-and-fro swing.
8	**(c)** kilometre per hour	km/h.
9	**(c)** distance-time graph	Plots distance vs time.
10	**(c)** oscillation	Bob returns to the starting point.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Distance travelled per unit time (1); m/s (1).	1+1
12	400/80 (1); 5 m/s (1).	1+1
13	Uniform: equal distances in equal time (1); Non-uniform: unequal (1).	1+1
14	Time for one complete oscillation (1); Measured in seconds (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	T = 60/30 = 2 s (1); Frequency = 30/60 (1); 0.5 per second (1).	1+1+1
16	72 x 5/18 = 20 m/s (1); 15 x 18/5 (1); 54 km/h (1).	1+1+1
17	20 min = 1/3 h (1); 18 x 1/3 (1); 6 km (1).	1+1+1
	OR: 180/4 = 45 km/h (1); 45 x 5/18 (1); 12.5 m/s (1).	1+1+1
18	400/5 = 80 km/h (1); 80 x 3 (1); 240 km (1).	1+1+1
	OR: Speedometer shows speed (1); Odometer shows distance (1); Helps check time and fuel (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Hang pendulums of lengths 25 cm and 100 cm (1); Time 20 oscillations each (1); Find T = t/20 (1); Longer has greater period (1); Mass has no effect (1).	1+1+1+1+1
	OR: Oscillation: one to-and-fro (1); Time period: time for one (1); Amplitude: extreme displacement (1); Bob returns (1); Periodic motion (1).	1+1+1+1+1
20	First 60 x 2 = 120 km (1); Second 40 x 3 = 120 km (1); Total 240 km in 5 h (1); Average 240/5 (1); 48 km/h (1).	1+1+1+1+1
	OR: Cyclist 5 m/s (1); Car 25 m/s (1); Snail 0.005 m/s (1); Arrange in order (1); Car fastest (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** distance / time	Speed.
2	**(c)** straight line	Constant speed gives a straight line.
3	**(d)** length	Longer pendulum, longer period.
4	**(b)** 60 km/h	120/2.
5	**(b)** at rest	Distance does not change.
6	**(c)** 2 s	40/20.
7	**(c)** 5 m/s	100/20.
8	**(d)** higher speed	More distance per time.
9	**(d)** the shadow of a rod	Shadow moves with the Sun.
10	**(d)** 180 km	60 x 3.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	54 x 5/18 (1); 15 m/s (1).	1+1
12	Sundial (1); Water clock (1).	1+1
13	One complete to-and-fro motion (1); Back to starting point (1).	1+1
14	Graph of distance against time (1); Shows motion (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hang a bob from a thread (1); Count 20 oscillations with a stopwatch (1); Divide the time by 20 (1).	1+1+1
16	Shadow of rod changes (1); Shows time of day (1); No sunlight at night (1).	1+1+1
17	Distance not proportional to time (1); Speed increases (1); Non-uniform (1).	1+1+1
	OR: Reduces error in starting and stopping (1); Larger time measured (1); Divide to get one (1).	1+1+1
18	Water flows at a steady rate (1); Level marks time (1); Measures fixed durations (1).	1+1+1
	OR: It increases (1); Longer pendulum swings slower (1); Mass does not matter (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sundial uses shadows (1); Water clock (1); Sand clock (1); Pendulum clock (1); Quartz watch (1).	1+1+1+1+1
	OR: Reset to zero (1); Start at the event (1); Stop at the end (1); Avoid reaction delay (1); Repeat and average (1).	1+1+1+1+1
20	Speed = distance / time (1); Uniform: straight line graph (1); Non-uniform: curved graph (1); Example: highway car (1); Example: bus in traffic (1).	1+1+1+1+1
	OR: 1 h = 60 min (1); 1 min = 60 s (1); 1 h = 3600 s (1); Common standard (1); Universal clocks (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 5/18 m/s	1000/3600.
2	**(c)** water clock	Clepsydra.
3	**(c)** 5 h	200/40.
4	**(d)** 10 m/s	36 x 5/18.
5	**(d)** quartz	Quartz crystals vibrate regularly.
6	**(c)** uniform	Constant speed.
7	**(b)** 72 km/h	20 x 18/5.
8	**(d)** swinging of a pendulum	Repeats regularly.
9	**(b)** non-uniform	Speed changes.
10	**(a)** m/s	Metre per second.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Its swings take equal time (1); Gives regular timing (1).	1+1
12	Speed = distance / time (1); 150 / 3 = 50 km/h (1).	1+1
13	Odometer: distance (1); Speedometer: speed (1).	1+1
14	Time = distance / speed (1); 90/45 = 2 h (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Uniform: equal distances in equal times (1); Non-uniform: not equal (1); Example: car on a highway vs in traffic (1).	1+1+1
16	Quartz crystal vibrates at a fixed rate (1); Counted by a circuit (1); Very accurate (1).	1+1+1
17	Sports (1); Railways (1); Experiments (1).	1+1+1
	OR: Straight line (1); Through origin (1); Slope shows speed (1).	1+1+1
18	One round in 60 s (1); So 60 s (1); Periodic motion (1).	1+1+1
	OR: Total distance 100 km (1); Total time 3 h (1); Average 33.3 km/h (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Equal distances in equal times (1); Uniform motion (1); Speed = 10/2 (1); 5 m/s (1); Graph is straight (1).	1+1+1+1+1
	OR: Distance 45 x 4 = 180 km (1); Time = 180/60 (1); 3 h (1); Speed up reduces time (1); Inversely proportional (1).	1+1+1+1+1
20	Odometer: distance (1); Speedometer: speed (1); Odometer from 12000 to 12120 km (1); Time 2 h (1); Average 60 km/h (1).	1+1+1+1+1
	OR: Moon's revolution (1); Pendulum (1); Heartbeat (1); Non-periodic: falling stone (1); Flying bird (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.